

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Total Marks:20

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I V.Sireesha(192472208) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V.Sireesha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well- explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

1.Smart Home Automation System Using Intelligent Agents

Introduction

A smart home automation system uses Artificial Intelligence (AI) to automatically control lighting, temperature, and security based on user behavior and environmental conditions. Intelligent agents collect information from sensors, process it, and perform suitable actions to improve comfort, energy efficiency, and security.

Simple Reflex Agent

A Simple Reflex Agent makes decisions only based on the current input without considering previous events.

Working in the Smart Home:

- Turns on lights when motion is detected.
- Switches off lights when no movement is detected.
- Turns on an alarm immediately if a door sensor detects intrusion.

Advantages

- Fast response.
- Easy to implement.
- Suitable for simple tasks.

Limitations

- Cannot remember previous states.
- Cannot learn user preferences.
- May make incorrect decisions in complex situations.

Model-Based Reflex Agent

A Model-Based Agent maintains an internal model of the environment and remembers previous states.

Working in the Smart Home:

- Remembers which rooms are occupied.
- Adjusts temperature based on indoor and outdoor conditions.
- Detects unusual activity by comparing current and previous sensor data.

Advantages

- Better decision-making.
- Works even with incomplete information.
- Improves system reliability.

Limitations

- Requires more memory.
- More complex than simple reflex agents.

Goal-Based Agent

A Goal-Based Agent selects actions that help achieve specific goals.

Working in the Smart Home:

- Maintains a comfortable room temperature.
- Ensures all doors are locked before bedtime.
- Reduces electricity consumption during peak hours.

Advantages

- Focuses on achieving objectives.
- Flexible in different situations.
- Makes intelligent decisions.

Limitations

- Requires planning.
- Decision-making takes more time.

Utility-Based Agent

A Utility-Based Agent chooses the action that provides the highest overall benefit.

Working in the Smart Home:

- Balances user comfort and electricity savings.
- Selects the most energy-efficient cooling option.
- Prioritizes security during emergencies while maintaining comfort.

Advantages

- Produces the best overall outcome.
- Handles multiple objectives.
- Optimizes system performance.

Limitations

- Requires utility calculations.
- More computationally expensive.

Comparison

Agent Type	Main Feature	Smart Home Example
Simple Reflex	Uses current input	Turns lights ON when motion is detected
Model-Based	Uses memory	Remembers occupied rooms
Goal-Based	Achieves goals	Maintains desired room temperature
Utility-Based	Maximizes overall benefit	Balances comfort, energy efficiency, and security

Best Choice

A Hybrid Agent combining Model-Based, Goal-Based, and Utility-Based approaches is the most effective.

Justification

- Learns user preferences over time.
- Maintains comfort automatically,saves electricity.
- Provides better security.

- Makes intelligent decisions using current and past information.

2.Robot Navigation in an Unknown Maze Using Uninformed Search

Introduction

A robot placed in an unknown maze must search for the exit without knowing the maze layout. Uninformed search algorithms explore the environment without using additional knowledge.

Uniform Cost Search (UCS)

Definition

Uniform Cost Search expands the node with the lowest path cost first.

Working

- Starts from the initial position.
- Calculates the cost of each path.
- Expands the least-cost path until the exit is found.

Advantages

- Finds the optimal path.
- Works well when path costs differ.
- Guarantees the lowest-cost solution.

Disadvantages

- Uses large memory.
- Slower for large search spaces.
- Time complexity increases with maze size.

Complexity

- **Time Complexity:** Exponential (depends on path cost)
- **Space Complexity:** High

Iterative Deepening Search (IDS)

Definition

Iterative Deepening Search repeatedly performs depth-limited searches with increasing depth limits until the goal is found.

Working

- Searches level by level.
- Increases search depth after each iteration.
- Stops when the exit is reached.

Advantages

- Requires less memory.
- Complete algorithm.
- Suitable for unknown search depth.

Disadvantages

- Repeats node expansion.
- Slightly slower because of repeated searches.

Complexity

- **Time Complexity:** $O(b^d)$
- **Space Complexity:** $O(bd)$

Comparison

Feature	UCS	IDS
Optimal Solution	Yes	Yes (equal step cost)
Memory Usage	High	Low
Speed	Slower	Faster in memory-limited systems
Suitable For	Different path costs	Unknown search depth

Relation to Intelligent Agents

Simple Reflex Agent

- Makes decisions only from current observations.
- Cannot efficiently solve an unknown maze.

Model-Based Agent

- Builds a map while exploring.
- Remembers visited paths.

Goal-Based Agent

- Plans actions to reach the exit.
- Selects paths that move toward the goal.

Utility-Based Agent

- Chooses the safest or least costly path.
- Optimizes overall performance.

Best Combination

The best solution is a Goal-Based Agent with Uniform Cost Search (UCS).

Justification

- Goal-Based Agent focuses on reaching the exit.
- UCS guarantees the least-cost path.
- Suitable for complex and unknown environments.
- Produces reliable and optimal navigation.

If memory is limited, Goal-Based Agent with Iterative Deepening Search (IDS) is also a good choice because IDS uses much less memory while still finding the solution.